Review_Comparison.pdf

GenAI Case Study: Customer Sentiment & Review Response Automation
Milestone 2 — Customer Segment Comparison & Prioritization

Business Question: "Which customers need immediate attention, and why?"

Dataset: 3,000 customer reviews, segmented into 3 groups of 1,000
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Executive Summary

Positive reviews are enthusiastic and detailed, celebrating product excellence. Neutral reviews are lukewarm and generic, indicating basic satisfaction without strong engagement. Negative reviews are highly emotional and urgent, demanding immediate resolution for significant product failures or dissatisfactions. While positive and negative express opposite sentiments, both tend to be driven by strong, specific experiences, whereas neutral reviews are often less descriptive and more indifferent.

(Generated by Gemini based on the segment-level metrics below — see Section 4.)

Segment	Review Count	Avg Sentiment Score	Avg Urgency Score	Avg Word Count	Top Distinctive Word
Positive (4-5★)	1,000	0.87	0.00	24.35	great
Neutral (3★)	1,000	0.42	0.00	21.14	nothing
Negative (1-2★)	1,000	-0.66	1.53	22.27	customer

1. Segment Overview

Reviews were segmented by star rating into three equal-sized groups of 1,000 reviews each: Positive (4-5★), Neutral (3★), and Negative (1-2★). This mirrors the same rating_bucket logic used in Milestone 1.

Conceptual Task 1: Why Negative Reviews Require Faster Escalation

The Negative segment has an average sentiment score of -0.66 and an average urgency score of 1.53 — meaning the typical negative review contains more than one and a half urgent-action keywords (e.g. defective, broken, refund, immediately). This is not just an emotional signal: Milestone 1's theme analysis showed these reviews are dominated by concrete, actionable language — product failure and direct requests for resolution. Unlike Neutral reviews, which express mild disappointment with no specific ask, Negative reviews represent customers who have already identified a problem and stated what they want done about it.

Business implication: Delaying a response to this segment carries compounding cost — an unresolved defective-product complaint can escalate into a chargeback, public review, or lost customer. This segment should be routed to support with the shortest possible response SLA.

Conceptual Task 2: Interpreting Urgency vs. Sentiment Differences

Sentiment score tells us how the customer feels (a smooth scale from -1 to +1); urgency score tells us how fast the business needs to act (a count of specific action-oriented keywords). They are related but not identical. In this dataset the

two aligned very cleanly — urgency is 0.00 for both Neutral and Positive and only nonzero for Negative — which is a useful confirmation of our keyword design, but also a characteristic of this templated dataset rather than a guarantee that sentiment and urgency will always move together. On organic real-world data, some Neutral or even Positive reviews could contain urgency language (e.g. “great product but I need a replacement part urgently”) — cases where both signals are needed to correctly triage a review.

2. Segment Metrics

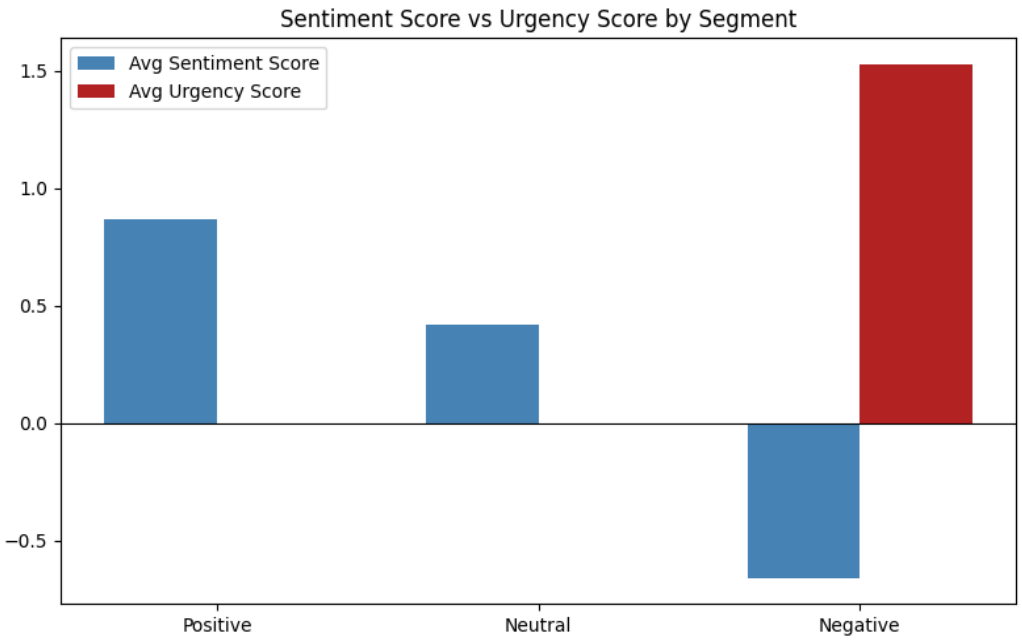


Figure 1: Average sentiment score vs. average urgency score by segment.

Sentiment cleanly separates the three groups (-0.66 → 0.42 → 0.87), while urgency is concentrated entirely in the Negative segment, reinforcing that Negative reviews combine both strongly negative tone and a concrete call to action.

3. Dominant Themes Across Segments

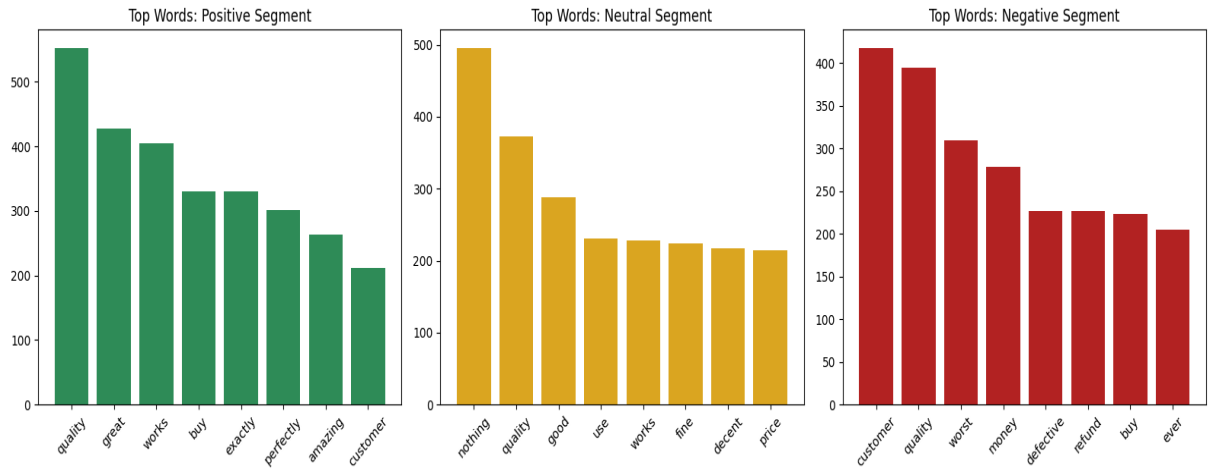


Figure 2: Top 8 words by segment (Positive / Neutral / Negative).

Category	Words
Common to all 3 segments	quality

Unique to Positive	exceeded, great, love, best, amazing, well, expectations, perfectly, purchase, highly, exactly
Unique to Neutral	okay, build, could, either, fair, good, fine, use, better, basic, decent, price
Unique to Negative	worst, money, refund, immediately, awful, want, work, ever, broken, terrible, defective

Conceptual Task 3: Recurring Product Issues Across Segments

Comparing top keywords across the three rating segments shows almost no vocabulary overlap — only “quality” appears across all three, confirming it is a generic, context-dependent word rather than a specific signal. The Positive segment is dominated by praise and superlative language; the Neutral segment has its own distinct identity of hedging, lukewarm language (“okay,” “fair,” “fine,” “decent,” “basic”) — not “diluted positive” or “diluted negative,” but genuinely ambivalent language; and the Negative segment centers on concrete product failure and direct action requests.

Business implication: The Neutral segment requires a different response strategy than Negative — these customers are not reporting defects, they are reporting an underwhelming (not broken) experience. A response system should not treat “okay” and “broken” the same way.

4. Numeric Comparison & GenAI Analysis

TF-IDF vectorization (via Scikit-learn) was used to turn each review into a numeric vector, downweighting generic words like “quality” and upweighting words distinctive to a smaller group. Each segment's average vector (“centroid”) was then compared using cosine similarity.

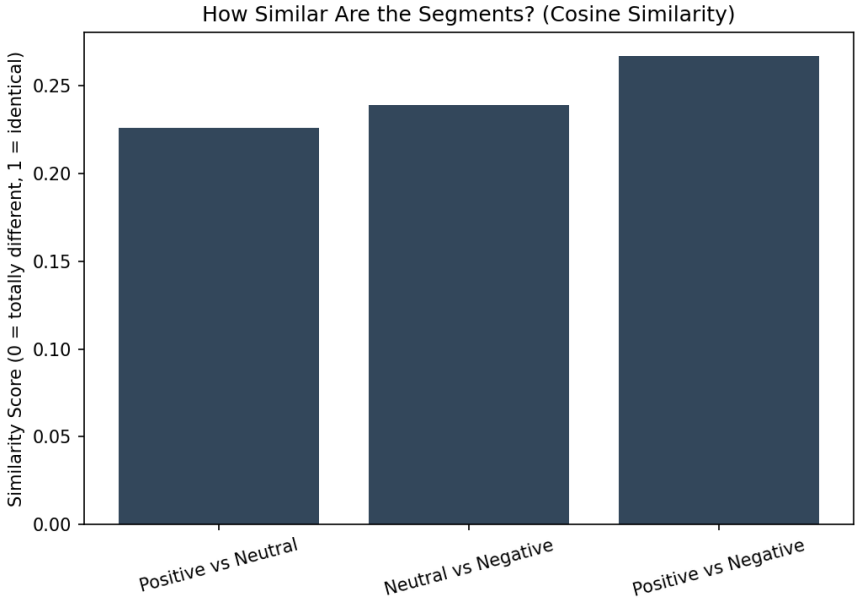


Figure 3: Cosine similarity between segment centroids (0 = totally different, 1 = identical).

All three similarity scores are low (0.23–0.27), confirming the segments are genuinely numerically distinct. A subtle, slightly counterintuitive result: Positive vs Negative (0.267) is marginally more similar than Positive vs Neutral (0.226), despite being opposite ends of the sentiment scale.

GenAI Explanation (via Gemini)

The slightly higher linguistic similarity between Positive and Negative reviews compared to Positive and Neutral is plausible because both extremes tend to be driven by strong emotional responses and specific, impactful experiences. Both very happy and very unhappy customers are highly motivated to express why they feel that way, often using more descriptive, active, and specific language (“works perfectly,” “exceeded expectations” vs. “worst product,” “defective”). This shared intensity of expression can lead to a similar underlying linguistic structure, even with opposite sentiment. Neutral reviews, by contrast, lack this strong emotional driver, using more generic language (“it's fine,” “decent,” “basic”), making them less similar to the highly descriptive language at both poles.

Segment “Personalities” (GenAI-Generated)

- **Positive:** These customers are your biggest fans — highly satisfied, enthusiastic, and often detailed about what they love. They're not looking for help, but expressing pure delight.
- **Neutral:** These customers are indifferent. The product is “fine” or “okay,” but it hasn't truly impressed them. They're not complaining, but not raving either.
- **Negative:** These customers are highly frustrated and demanding immediate action, often referring to defective products and wanting a refund. Their language is emotional, urgent, and critical.

Recommended Actions per Segment (GenAI-Generated)

- **Positive — Leverage and Learn:** Thank these customers, identify what drives satisfaction, and encourage testimonials/referrals. This segment is the brand advocacy base.
- **Neutral — Probe and Improve:** Proactively engage to understand “why just okay,” using targeted follow-ups. This segment is ripe for engagement and potential uplift into loyalty.
- **Negative — Prioritize and Resolve Immediately:** Given the high urgency score and intense language, these reviews demand immediate support intervention — refund, replacement, or troubleshooting — and should inform systemic product/QA fixes.

5. Segmentation Value & Limitations

Conceptual Task 4: How Segmentation Supports Decision-Making

Treating all 3,000 reviews as one undifferentiated pool would force a single, generic response strategy — which the data shows would be wrong for at least two of the three segments. Segmentation converts one large, ambiguous dataset into three groups with distinct, measurable characteristics, each implying a different action: Positive is a low-effort advocacy opportunity; Neutral is an under-served opportunity segment worth proactive outreach; Negative is a resourcing and escalation priority that should determine support SLA and staffing.

Without segmentation, a single blended average-sentiment metric (roughly 0.21 across all reviews) would mask both the churn risk in the Negative segment and the missed opportunity in the Neutral segment. Segmentation directly supports resource allocation, product prioritization, and marketing decisions — each segment maps to a different team and action, which a single blended score cannot provide.

Conceptual Task 5: Potential Bias in AI-Driven Segmentation

- **Rating-based segmentation bias:** Segmenting primarily by star rating doesn't perfectly track actual sentiment — Milestone 1 showed 78% of Neutral (3-star) reviews are text-Positive. Rating alone can misclassify satisfied customers as “needs improvement,” and can miss a smaller genuinely unhappy 3-star group.
- **Keyword-selection bias in urgency scoring:** The urgency word list was derived by looking at what showed up in negative reviews, meaning it was somewhat fit to this dataset. On real-world data, this could miss urgent issues phrased with different vocabulary.
- **Sentiment-tool bias:** VADER is rule-based and tuned for general English text; it can misread sarcasm, industry-specific language, or culturally-specific phrasing, systematically under- or over-scoring some customers.
- **Synthetic/templated data bias:** This dataset shows templated review structures. Patterns found here (e.g. “quality” appearing everywhere, urgency = 0 outside Negative) may be dataset artifacts rather than genuine properties of real customer language, and would need re-validation on organic review data before production use.

Business implication: AI-driven segmentation should be treated as a decision-support tool, not an autonomous decision-maker — outputs should be periodically audited against human judgment (especially the Neutral segment, where misclassification risk is highest), and keyword lists revisited regularly as new complaint language emerges.

6. Summary & Priority Recommendations

- **Negative (1-2★):** Highest priority for support SLA — combines strongly negative sentiment (-0.66) with a real urgency signal (1.53) and explicit refund/resolution requests.

• **Neutral (3★):** A distinct, under-served opportunity segment with its own “lukewarm” vocabulary — worth proactive outreach to convert to loyalty, not just deprioritized as “average.”

• **Positive (4-5★):** Lowest support burden, highest advocacy value — route to testimonial/referral requests.

• Segmentation quantitatively confirms all three groups are numerically distinct (cosine similarity 0.23–0.27 between all pairs), validating that a single blended metric would obscure real, actionable differences between customer groups.

Note: As in Milestone 1, this dataset's templated structure means urgency and similarity findings should be validated against organic review data before being used to design a production triage system.